Problem 3. For the questions 10,11,12 consider the circuit shown in figure 2 and answer using nodal analysis:

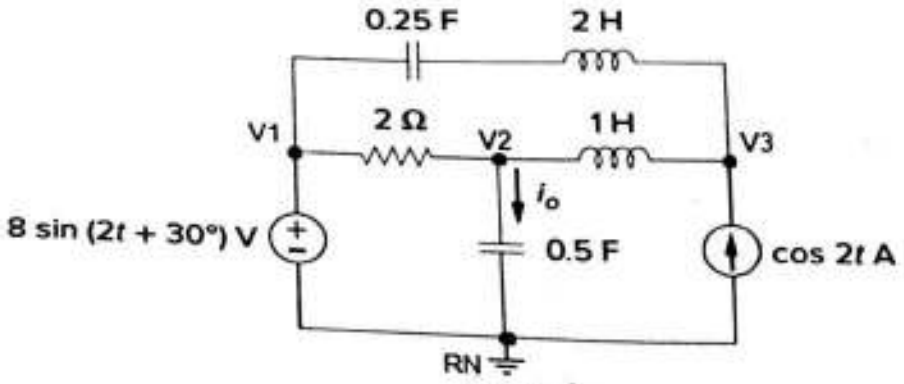


Fig. 2

10) The voltage (V1) in phasor domain is:

.....

.....

11) Write the required nodal equations (do not simplify):

.....

.....

.....

Problem 4. For the questions 13,14,15 consider the circuit shown in figure 3 and answer:

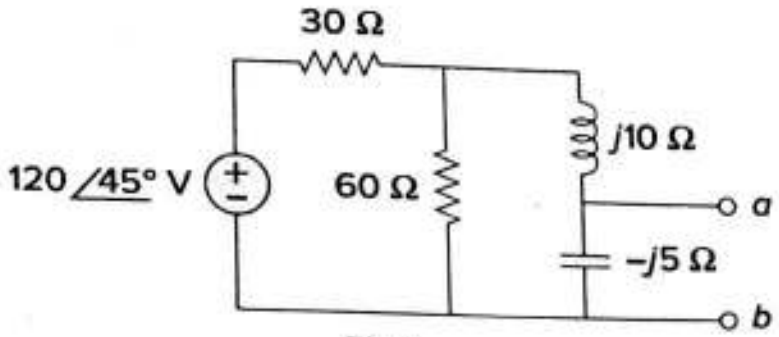


Fig. 3

12) The Thevenin impedance (Z_{Th}) with respect to the terminals a-b is:

.....

13) The Thevenin voltage (V_{Th}) with respect to the terminals a-b is:

.....

.....

Problem 1. For the questions 1,2,3,4,5,6 consider:

- $v(t)= 500\cos(1000t+30^{\circ})$ and $i(t)= -20\sin(1000t-120^{\circ})$.

1) The voltage leads the current in degrees by:

.....

2) The current lags the voltage in radians by:

.....

3) The voltage value (v) at t=1ms is:

.....

4) The phasor domain representation of i(t) is:

.....

5) The phasor domain representation of v(t) is:

.....

6) The frequency of the voltage in Hz is:

.....

Problem 2. For the questions 7,8,9 consider the circuit shown in figure 1 and answer:

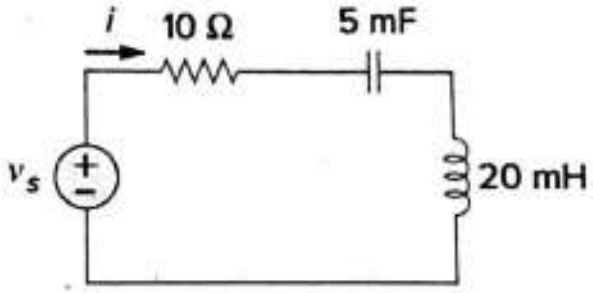


Fig.1

- $v_s(t) = 100\cos(200t + 45^0)$ V

7) The equivalent impedance at the terminals of the voltage source in (Ω) is:

.....

8) The admittance at the terminals of the voltage source in (S) is:

.....

9) The steady state current (i) in time domain is:

.....